

PHYS 253: Thermal Physics

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Abstract

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1 Introduction

1.1 Course Information

The textbook is *An Introduction to Thermal Physics* by Daniel V. Schroeder. Grading is divided as follows:

- 10% problem sets (~ 6 , submitted as handwritten PDF scans)¹
- 50% two midterms
- 40% final exam

2 Thermodynamics

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Definition 1 (Thermodynamics). *Thermodynamics, from the ancient Greek $\theta\epsilon\rho\mu\epsilon$ (heat) and $\delta\upsilon\nu\alpha\mu\iota\varsigma$ (power), is the branch of physics that deals with the relationships between heat, work, temperature and energy. It studies the equilibrium properties of large-scale systems in which temperature is an important variable.*

Remark 1. *Thermodynamics does not describe the microscopic details of a system². It describes the macroscopic properties and behaviour of a system using a small number of state variables.*

2.1 Terminologies

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Definition 2 (System, surroundings, boundary). *The system is everything involved in the process under study; the surroundings are the rest of the universe it interacts with; the boundary is the interface between them.*

Definition 3 (Open, closed, isolated). *An open system exchanges matter and energy with its surroundings, a closed system only energy, and an isolated system neither. The boundary determines which.*

2.1.1 Equilibrium States

Definition 4 (Equilibrium state). *A system is in an equilibrium state if all the measurable bulk physical properties of the system are uniform throughout the system and constant in time.*

Remark 2. *For a fixed volume of gas in an equilibrium state, the pressure is the same everywhere in the gas and the volume does not change in time.*

Remark 3. *The pressure, volume and mass of a gas fully describe its equilibrium state. Once those are fixed, every other measurable bulk property is fixed as well.*

¹Only one or two problems on each assignment are marked for submission and grading.

²Statistical mechanics, and more recent extensions such as stochastic thermodynamics, do.

2.1.2 Equations of State

Definition 5 (Equation of state). *An equation of state is a functional relationship between the easily measured state variables³ of a system in an equilibrium state.*

Example. *The ideal gas law is an equation of state relating (P, V, T) . It applies to equilibrium states of dilute gases at high temperature:*

$$PV = nRT = Nk_B T,$$

where n is the number of moles, $N = nN_A$ is the number of molecules, R is the ideal gas constant and $k_B = R/N_A = 1.38 \times 10^{-23} \text{ J K}^{-1}$ is Boltzmann's constant.

2.1.3 State Functions

Definition 6 (State function). *A state function is a property of a system that depends only on the current equilibrium state, and not on the path taken to reach that state. Internal energy, enthalpy, entropy and Gibbs free energy are all state functions.*

Remark 4. *State functions matter because their values can be compared between two equilibrium states without knowing anything about the process that connected them.*

Example. *Temperature is a state function. It is the property that determines whether two systems are in thermal equilibrium⁴ with each other.*

3 Laws of Thermodynamics

3.1 Zeroth Law

Law 1 (Zeroth Law of Thermodynamics). *If two systems A and B are each separately in thermal equilibrium with a third system C , meaning no heat flows between A and C and no heat flows between B and C , then A and B are in thermal equilibrium with each other.*

Remark 5. *The zeroth law is what makes a thermometer work. System C is the thermometer, and the law guarantees that two systems reading the same value on it are in thermal equilibrium with each other.*

Remark 6. *Thermal equilibrium (equal T) is not full thermodynamic equilibrium, which also requires mechanical and chemical (diffusive) equilibrium.*

Remark 7. *Extrapolating $V(T)$ at fixed P for any sample of ideal gas gives $V \rightarrow 0$ at the same temperature, $-273.15^\circ \text{C} = 0 \text{ K}$, which defines absolute zero. A gas thermometer reads $T = PV/nR$.*

³Such as pressure, volume and temperature.

⁴Thermal equilibrium only, not full thermodynamic equilibrium.

3.2 First Law

Law 2 (First Law of Thermodynamics). *The relationship between the heat Q added to a system, the work W done on it and the change in its internal energy U is dictated by the conservation of energy:*

$$\Delta U = Q + W.$$

Remark 8. *Sign convention: $Q > 0$ when heat flows into the system, and $W > 0$ when work is done on the system. Both raise the internal energy.*

Remark 9. *For an isolated system $Q = W = 0$, so U is constant.*

4 Energy, Work and Heat

4.1 Energy

Definition 7 (Energy). *Energy, from the Greek *energeia* meaning "in work," is the capacity to do work. It is a property of a thermodynamic system, and it is stored in different forms depending on the phase and composition of the matter involved.*

Forms of energy.

Definition 8 (Kinetic energy). *From the Greek *kinesis* (motion), the energy associated with the motion of a body or of the particles inside it.*

Definition 9 (Potential energy). *Energy due to condition, position or composition. It is associated with interactions, meaning forces of attraction or repulsion between objects, or chemical bonding.*

Definition 10 (Internal energy of an ideal gas). *For an ideal gas the internal energy depends only on temperature. It is the number of molecules times the average kinetic energy per molecule:*

$$U(T) = N\bar{\epsilon}_k = \frac{f}{2}Nk_B T,$$

where f is the number of degrees of freedom per molecule. For a monatomic gas $f = 3$ (translation only), so $U = \frac{3}{2}Nk_B T$. More complex molecules have extra rotational or vibrational degrees of freedom, so $f > 3$.

4.2 Work

Definition 11 (Work). *Work is done when a force acts through a distance, and it is associated with a change in the energy of a system:*

$$W = \int \vec{F} \cdot d\vec{r}.$$

In thermodynamics, work is any transfer of energy into or out of a system that is not heat.

Compression of a gas by a piston. Take a gas in a cylinder closed by a piston of area A . Pushing the piston in by a small distance δx does work

$$W = F \delta x.$$

The pressure of the gas is $P = F/A$, so $F = PA$ and

$$W = PA \delta x.$$

For a quasistatic compression⁵, the change in volume is $\delta V = -A \delta x$, so the work done *on* the gas is

$$W = -P \delta V \quad (\text{quasistatic}).$$

Compressing the gas means $\delta V < 0$, so $W > 0$ and the energy of the gas goes up, as expected. For a finite change in volume the work is the integral

$$W = - \int_{V_i}^{V_f} P dV,$$

which is the area under the curve on a P - V diagram.

Remark 10. *Energy and work are measured in the same units, joules, where $1 \text{ J} = 1 \text{ N m} = 1 \text{ kg m}^2 \text{ s}^{-2}$.*

4.3 Heat

Definition 12 (Heat). *Heat is energy that flows between systems because of a temperature difference, and it drives them toward thermal equilibrium. It is the quantity of energy Q transferred between a system and its surroundings as a result of that temperature difference. Heat flows spontaneously from the warmer body to the colder one, by conduction, convection or radiation, until $T_A = T_B$. No temperature difference, no heat flow.*

Remark 11. *Heat and work are not properties of a thermodynamic system. They are the two ways energy can be exchanged between a system and its surroundings. A system does not "contain" heat or work, it contains energy.*

5 Compression and Expansion of an Ideal Gas

5.1 The Two Limiting Cases

The expression $W = -P \delta V$ is deceptively simple. For an ideal gas $PV = nRT$, so P and V are not independent state variables: how P changes as V changes depends on what happens to T during the process. Two limiting cases cover most situations.

⁵Slow enough that the gas stays in internal equilibrium at every instant, so the pressure is uniform throughout the volume and the ideal gas law still applies.

- Isothermal: T is constant during the process. The gas stays in contact with a reservoir that absorbs or supplies whatever heat is needed.
- Adiabatic: no exchange of heat during the process, meaning no heat flows through the boundary. The gas is insulated, or the process is too fast for heat to flow.

Both are assumed quasistatic, so $PV = nRT = Nk_B T$ holds at every instant.

5.2 Quasistatic Isothermal Compression and Expansion

Work. Since T is constant, $P = Nk_B T/V$ can be substituted directly into the work integral:

$$W = - \int_{V_i}^{V_f} \frac{Nk_B T}{V} dV = -Nk_B T \ln \frac{V_f}{V_i} = Nk_B T \ln \frac{V_i}{V_f}.$$

For a compression $V_f < V_i$, so $W > 0$: work is done on the gas. For an expansion $V_f > V_i$ and $W < 0$: the gas does work on its surroundings.

Heat. For an ideal gas U depends only on T , so an isothermal process has $\Delta U = 0$. The first law then gives

$$Q = -W = Nk_B T \ln \frac{V_f}{V_i}.$$

In an isothermal expansion the gas does work but its energy does not drop, so it must absorb an equal amount of heat from the reservoir. In an isothermal compression the work done on the gas is dumped into the reservoir as heat.

5.3 Quasistatic Adiabatic Compression and Expansion

There is no heat, so the first law reduces to

$$\Delta U = Q + W = 0 + W = W.$$

Any work done on the gas goes entirely into its internal energy, and since $U = \frac{f}{2} Nk_B T$, the temperature must change: an adiabatic compression heats the gas and an adiabatic expansion cools it.

Equation of the adiabat. Write the first law for an infinitesimal step and use $dW = -P dV$:

$$dU = \frac{f}{2} Nk_B dT = -P dV.$$

Substitute $P = Nk_B T/V$ from the ideal gas law and cancel Nk_B :

$$\frac{f}{2} \frac{dT}{T} = -\frac{dV}{V}.$$

Integrate from the initial to the final state:

$$\frac{f}{2} \ln \frac{T_f}{T_i} = - \ln \frac{V_f}{V_i}.$$

Exponentiating both sides gives

$$V_f T_f^{f/2} = V_i T_i^{f/2},$$

so $VT^{f/2}$ is constant along an adiabat.

Proposition 1 (Adiabat in terms of P and V). *Eliminating T with the ideal gas law $T = PV/Nk_B$ gives*

$$PV^\gamma = \text{constant}, \quad \gamma = \frac{f+2}{f},$$

where γ is called the *adiabatic exponent*.

Remark 12. Since $\gamma > 1$, an adiabat is steeper than an isotherm ($PV = \text{constant}$) on a P - V diagram. For a monatomic gas $f = 3$, so $\gamma = 5/3$.

6 Heat Capacity

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6.1 Definition

Definition 13 (Heat capacity). *The heat capacity C of a system is the amount of heat Q required to raise its temperature by $\Delta T = 1 \text{ K}$ (or 1°C):*

$$C = \frac{Q}{\Delta T}.$$

It is an extensive quantity: it depends on the size of the system. Doubling the amount of material doubles the heat capacity.

Definition 14 (Specific heat capacity). *The specific heat capacity is the heat capacity per unit mass,*

$$c \equiv \frac{C}{m}.$$

It is an intensive quantity and a fundamental property of the material.

6.2 Heat Capacity Depends on the Process

Using the first law, $Q = \Delta U - W$, so

$$C = \frac{Q}{\Delta T} = \frac{\Delta U - W}{\Delta T}.$$

This exposes an ambiguity in the definition: C depends on how much work is done during the heating. In this sense C would almost be better named the "energy capacity," since it does not really matter to the system whether the energy enters as Q or as W . To make C well defined we must specify the process, and in particular whether $W = 0$ or $W \neq 0$.

Definition 15 (Heat capacity at constant volume). *If the volume is held fixed, then $W = -P \Delta V = 0$ and all the heat goes into internal energy:*

$$C_V = \left(\frac{\Delta U}{\Delta T} \right)_V = \left(\frac{\partial U}{\partial T} \right)_V.$$

Most materials expand as they are heated (as U increases). This is most obvious for gases, but solids do it too⁶. Unless the volume is deliberately fixed, the system expands against its surroundings as it is heated, which means it does work on the surroundings: $W < 0$. Additional heat is then needed to compensate for that lost energy, so the heat capacity is larger than C_V .

Definition 16 (Heat capacity at constant pressure). *Typically the expansion happens at constant pressure, namely the pressure exerted by the surroundings, which is often atmospheric pressure. Then*

$$C_P = \left(\frac{\Delta U - (-P \Delta V)}{\Delta T} \right)_P = \left(\frac{\partial U}{\partial T} \right)_P + P \left(\frac{\partial V}{\partial T} \right)_P.$$

Remark 13. *The first term is not exactly C_V : the partial derivative is taken at constant pressure, not constant volume.*

Remark 14. *The second term is the additional heat needed to compensate for the energy lost as work. The more the volume increases, the larger this term is. For solids and liquids $\partial V / \partial T \approx 0$, so it can often be neglected and $C_P \approx C_V$. For gases it is significant.*

Proposition 2 (C_P of an ideal gas). *For an ideal gas U depends only on T , so $(\partial U / \partial T)_P = C_V$, and $V = Nk_B T / P$ gives $(\partial V / \partial T)_P = Nk_B / P$. Hence*

$$C_P = C_V + Nk_B = C_V + nR, \quad c_P = c_V + R \text{ (molar)}.$$

6.3 Measuring Specific Heat: Calorimetry

The best way to determine C or c is to measure it, using a calorimeter: an insulated container in which two bodies at different temperatures are brought into thermal contact. Since the whole system is isolated, conservation of energy gives

$$Q_{\text{system}} = -Q_{\text{surroundings}},$$

and once thermal equilibrium is reached both bodies are at the same final temperature.

Example. *A 150.0 g sample of lead is heated to 100 °C and dropped into 50.0 g of water at 22.0 °C in an insulated container. The final temperature is 28.8 °C. Find the specific heat of lead, given that the specific heat of water is well known, $c_{\text{water}} \approx 4.18 \text{ J g}^{-1} \text{ K}^{-1}$.*

⁶Aside: some materials contract as they are heated over certain ranges of T (negative thermal expansion), but they are the exception.

Take the lead as the system and the water as the surroundings, so $q_{\text{lead}} + q_{\text{water}} = 0$. The heat absorbed by the water is

$$q_{\text{water}} = mc \Delta T = (50 \text{ g})(4.18 \text{ J g}^{-1} \text{ }^\circ\text{C}^{-1})(28.8 - 22.0)^\circ\text{C} = 1.4 \times 10^3 \text{ J}.$$

Then $q_{\text{lead}} = -q_{\text{water}} = -1.4 \times 10^3 \text{ J}$, and since $q_{\text{lead}} = m_{\text{lead}} c_{\text{lead}} \Delta T_{\text{lead}}$,

$$c_{\text{lead}} = \frac{-1.4 \times 10^3 \text{ J}}{(150 \text{ g})(28.8 - 100)^\circ\text{C}} = 0.13 \text{ J g}^{-1} \text{ }^\circ\text{C}^{-1}.$$

6.4 The Mechanical Equivalent of Heat

James Joule (1818–1889) is best known for an experiment in which a falling weight spins a paddle wheel inside an insulated barrel of water. The weight loses potential energy $mg \Delta h$, and Joule measured the resulting rise in the temperature of the water. In 1850 he found that about 4.2 J of mechanical energy was needed to raise the temperature of 1 g of water by 1 $^\circ\text{C}$, quite close to the twentieth-century value of the specific heat of water, $4.18 \text{ J g}^{-1} \text{ K}^{-1}$. The experiment established two things:

- W and Q are mutually interchangeable: energy can change form.
- A given amount of work always generates the same amount of heat, provided the work is totally converted to heat: energy is conserved.

Remark 15. See Problem 1.30 in Schroeder, and try it: put a few spoonfuls of room-temperature water in a bottle with a tight lid, measure the temperature, shake it as hard as you can for several minutes, measure again, and compare with a rough calculation of the expected temperature change.

7 Thermodynamic Processes

7.1 Heat and Work Along a Path

Heat and work are not state functions, so their infinitesimal amounts are written with an inexact differential, dQ and dW , to signal that they depend on the path and not just on the endpoints. The totals for a process are obtained by integrating along the path:

$$W = \int dW = - \int_{V_1}^{V_2} P dV, \quad Q = \int dQ.$$

On a P - V diagram the work is the area under the path, so different paths between the same two states give different W and Q , even though $\Delta U = Q + W$ is the same for all of them.

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7.2 The Four Basic Processes

Figure 1 shows the four standard quasistatic processes of an ideal gas on a P - V diagram, together with two cycles built from them.

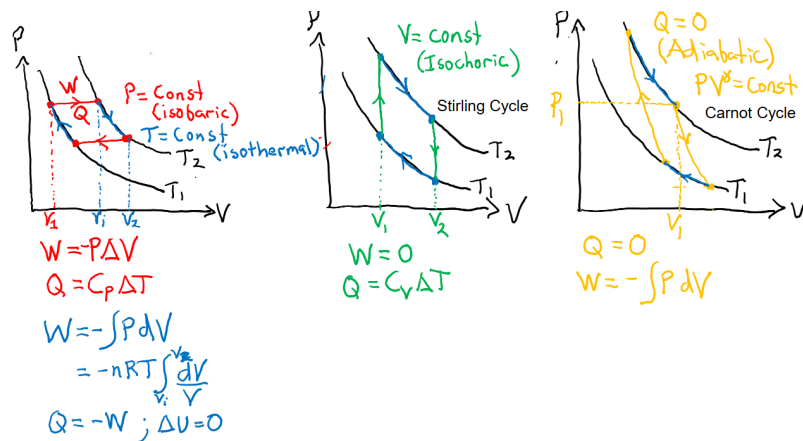


Figure 1: Left: isobaric (red) and isothermal (blue) processes between the isotherms T_1 and T_2 . Centre: a Stirling cycle, made of two isochoric (green) and two isothermal steps. Right: a Carnot cycle, made of two adiabatic (yellow) and two isothermal steps.

Isobaric ($P = \text{constant}$). The path is a horizontal line. The work is simply

$$W = -P \Delta V = -P(V_2 - V_1),$$

and the heat is set by the heat capacity at constant pressure:

$$dQ = C_P(T) dT, \quad Q = \int_{T_1}^{T_2} C_P(T) dT = C_P \Delta T = C_P(T_2 - T_1) \quad (\text{if } C_P \text{ is constant}).$$

Isothermal ($T = \text{constant}$). The path follows an isotherm $PV = nRT$. Using $P = nRT/V$,

$$W = - \int_{V_1}^{V_2} P dV = -nRT \int_{V_1}^{V_2} \frac{dV}{V} = -nRT \ln \frac{V_2}{V_1}.$$

Since U depends only on T for an ideal gas, $\Delta U = 0$ and so $Q = -W$.

Isochoric ($V = \text{constant}$). The path is a vertical line, so no work is done:

$$W = 0, \quad Q = \Delta U = C_V \Delta T.$$

Adiabatic ($Q = 0$). No heat flows, and the path follows an adiabat $PV^\gamma = \text{constant}$, which is steeper than the isotherms. All the energy change is work:

$$Q = 0, \quad W = - \int_{V_1}^{V_2} P dV = \Delta U.$$

7.3 Cycles

Definition 17 (Heat engine). *A system that converts thermal energy into work by taking a working substance through a cycle between a hot reservoir at T_2 and a cold one at T_1 .*

A cycle is a closed path on the P - V diagram: the system returns to its initial state, so $\Delta U = 0$ over one cycle and $Q = -W$. The net work is the area enclosed by the loop.

- Stirling cycle: two isotherms (T_1, T_2) joined by two isochores (V_1, V_2).
- Carnot cycle: two isotherms (T_1, T_2) joined by two adiabats.

8 Kinetic Theory of Gases

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Kinetic theory is a microscopic model for $U(T)$ and C of an ideal gas. The plan has two steps:

1. Determine how the pressure P is related to the kinetic energy of the gas particles. This can be worked out for a single particle and then generalized to all N .
2. Use the ideal gas equation of state to relate the result to T .

8.1 Pressure from a Single Particle

Momentum is conserved in the elastic collisions between gas particles and the walls of the container. The pressure is the force exerted by the particles on a wall, and by Newton's second law

$$F = ma = \frac{dp}{dt},$$

where p is the momentum.

Consider one particle of mass m bouncing back and forth between the left and right walls of a box of length L , with x -velocity u_x . A graph of the force on the left wall against time shows a series of spikes, one per collision, separated by the round-trip time

$$\Delta t = \frac{2L}{u_x}.$$

At each collision the particle's x -momentum reverses, from mu_x to $-mu_x$. By conservation of momentum, the momentum given to the wall is

$$\Delta p_{\text{wall}} = -\Delta p_{\text{gas}} = -(p_f - p_i) = -(-mu_x - mu_x) = 2mu_x.$$

The average force is the momentum transferred per unit time, so $\Delta p_{\text{wall}} = \bar{F} \Delta t$, that is

$$2mu_x = \bar{F} \cdot \frac{2L}{u_x} \implies \bar{F} = \frac{mu_x^2}{L} \quad (\text{one particle}).$$

8.2 Pressure from All N Particles

Each particle contributes its own mu_x^2/L , so summing over all N particles and writing $\overline{u_x^2}$ for the average of u_x^2 over the gas,

$$\bar{F} = N \frac{m\overline{u_x^2}}{L}.$$

Dividing by the wall area A , and using $AL = V$,

$$P = \frac{\bar{F}}{A} = \frac{Nm\overline{u_x^2}}{V} \implies PV = Nm\overline{u_x^2}.$$

8.3 Temperature and the Average Kinetic Energy

The velocity has no preferred direction, so the three axes contribute equally to the mean square speed:

$$\overline{u^2} = \overline{u_x^2} + \overline{u_y^2} + \overline{u_z^2} = 3\overline{u_x^2}.$$

Substituting $\overline{u_x^2} = \frac{1}{3}\overline{u^2}$ into $PV = Nm\overline{u_x^2}$ gives

$$PV = \frac{1}{3}Nm\overline{u^2}.$$

Comparing with the ideal gas law $PV = Nk_B T$ and cancelling N ,

$$k_B T = \frac{1}{3}m\overline{u^2} \implies \bar{e}_k = \frac{1}{2}m\overline{u^2} = \frac{3}{2}k_B T.$$

Remark 16. *The average translational kinetic energy per molecule depends only on temperature, not on the mass or type of molecule. This is the microscopic meaning of temperature.*

Remark 17. *The molecular speeds follow a distribution (Maxwell-Boltzmann), with most probable speed $<$ average speed $<$ rms speed $\sqrt{\overline{u^2}}$.*

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9 Equipartition and Degrees of Freedom

9.1 The Equipartition Theorem

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The result $\bar{e}_k = \frac{3}{2}k_B T$ is a special case of a far more general statement. It applies to every form of energy that is a quadratic function of a coordinate or velocity component (Cartesian or angular). Each such term is called a degree of freedom.

Definition 18 (Degree of freedom). *A degree of freedom (DOF) is a form of energy that enters the total energy as a quadratic function of a position or velocity component. Translational kinetic energy, rotational kinetic energy, vibrational kinetic energy and elastic potential energy are all quadratic degrees of freedom.*

Principle 1 (Equipartition theorem). *In a system at thermal equilibrium at temperature T , each accessible quadratic degree of freedom carries an average energy*

$$\langle e \rangle = \frac{1}{2}k_B T.$$

For N molecules, each with f accessible degrees of freedom and no other temperature-dependent energy, the total thermal energy is

$$U_{\text{thermal}} = Nf \cdot \frac{1}{2}k_B T.$$

Remark 18. *Except for ideal gases, U_{thermal} is never the total energy of a system: there is also "static" energy that does not change with temperature, such as chemical bond energy and the rest energies mc^2 of the particles. Equipartition remains a good approximation for energy changes associated with raising or lowering the temperature, provided no phase transformations or bond-breaking reactions occur over the range considered.*

9.2 Counting Degrees of Freedom

Consider a molecule of N atoms. Specifying the positions of all atoms takes $3N$ coordinates, so there are $3N$ independent displacements. These split into three physical types.

- **Translational:** three coordinates fix the centre of mass, so 3 displacements are translations of the molecule as a whole.
- **Rotational:** of the remaining $3N - 3$ internal displacements, orienting the molecule takes two angles for a linear molecule (the direction of the axis) and three for a nonlinear one (the extra angle fixes rotation about that axis). So 2 (linear) or 3 (nonlinear) are rotations.
- **Vibrational:** what remains is the atoms moving relative to one another,

$$N_{\text{vib}} = 3N - 5 \text{ (linear)}, \quad 3N - 6 \text{ (nonlinear)}.$$

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Remark 19. *Translational and rotational modes count as one quadratic DOF each (kinetic energy only); each vibrational mode counts as two (kinetic and potential). So a fully excited nonlinear molecule has $f = 3 + 3 + 2(3N - 6)$.*

Example (NH_3). $N = 4$, *nonlinear*: $3N = 12$ coordinates, 3 translational, 3 rotational, so $12 - 6 = 6$ vibrational modes and $f_{\text{vib}} = 12$. Total $f = 3 + 3 + 12 = 18$.

9.3 Degrees of Freedom in Solids

In a crystal the atoms are pinned near lattice sites, so the only motion is vibration, and each vibrational mode counts twice (KE and PE).

- **Simple solid** (one atom per unit cell): each atom vibrates along 3 independent axes, each vibration being 2 quadratic DOF, so

$$f = f_{\text{vib}} = 3 \times 2 = 6.$$

- **Complex solid** (n atoms per unit cell): there are $3n$ vibrational displacements per cell, each with 2 DOF, so

$$f = f_{\text{vib}} = 3n \times 2 = 6n.$$

9.4 Freezing Out

Not every degree of freedom contributes at every temperature. Each DOF has a minimum quantum excitation energy (for a vibration, $\hbar\omega_{\text{vib}}$). If

$$\frac{1}{2}k_B T \ll (\text{minimum excitation energy}),$$

there is not enough thermal energy to lift the DOF out of its ground state. The DOF is then frozen out: inaccessible, with no thermal energy flowing into it. The effective number of degrees of freedom therefore depends on temperature, $f \rightarrow f(T)$.

10 Predicting Heat Capacities

10.1 Ideal Gas

From equipartition $U = \frac{f}{2}Nk_B T$, so the heat capacity at constant volume is

$$C_V = \left(\frac{\partial U}{\partial T} \right)_V = \frac{\partial}{\partial T} \left(\frac{f}{2}Nk_B T \right) = \frac{f}{2}Nk_B.$$

When degrees of freedom freeze out below their threshold temperatures, $f \rightarrow f(T)$ and hence $C_V \rightarrow C_V(T)$.

Example. In a monatomic gas such as helium, $f = 3$ (translation only), and these cannot freeze out. So

$$C_V = \frac{3}{2}Nk_B = \frac{3}{2}nR, \quad R = 8.31 \text{ J mol}^{-1} \text{ K}^{-1}.$$

Per mole,

$$c_V = \frac{C_V}{n} = \frac{3}{2}R \approx 12.5 \text{ J mol}^{-1} \text{ K}^{-1},$$

independent of the type of atom and of temperature.

Example (Diatomic H_2). As T rises, degrees of freedom unfreeze one type at a time: $c_V = \frac{3}{2}R$ at low T (translation, $f = 3$), $\frac{5}{2}R$ once rotation turns on ($f_{\text{rot}} = 2$, above $\sim 100 \text{ K}$), and $\frac{7}{2}R$ once vibration turns on ($f_{\text{vib}} = 2$, when $k_B T \sim \hbar \omega_{\text{vib}}$, thousands of K).

10.2 Solids: the Rule of Dulong and Petit

At sufficiently high temperature, with all vibrational modes active and none frozen out, the molar heat capacity of a solid tends to

$$C_V \sim \frac{f_{\text{vib}}}{2}R.$$

For a monatomic solid $f_{\text{vib}} = 6$, giving

$$C_V \sim 3R \approx 25 \text{ J mol}^{-1} \text{ K}^{-1}.$$

11 Thermodynamic Equilibrium Revisited

11.1 Local Thermodynamic Equilibrium

Earlier an equilibrium state was described as one whose easily measurable macroscopic properties are constant in time and homogeneous throughout the system. This can be relaxed so that thermodynamic concepts apply to systems not yet in complete equilibrium.

Definition 19 (Local thermodynamic equilibrium). *Divide the system into small volume elements. A volume element V_i is in local thermodynamic equilibrium if the easily measurable macroscopic properties inside V_i match the true equilibrium values for the local (P, V, T) conditions, that is, if the properties within V_i are the same as those of an equilibrium system at those local conditions.*

Remark 20. *This lets a system with, say, a temperature gradient still be treated cell by cell: each cell is locally in equilibrium even though the system as a whole is not. The approximation is better when relaxation is slow and the gradients in (P, V, T) are gentle. Within each cell one can then talk about U , H , S .*

11.2 Equilibrium versus Rates

To determine the equilibrium state of a system we usually need not worry about how long it takes to get there. Thermodynamics, by most definitions, is the study of equilibrium states themselves. Questions about time and the rates of processes belong to a separate but related subject, sometimes called transport theory or kinetics.

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A Appendix

B Useful Links